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LECTURE 1

Random Variables

1.1 Bogleheads Own the Entire Market

- John Bogle popularized the index fund and a simple investment philosophy: “Own the entire market”
- Followers of this philosophy affectionately call themselves “Bogleheads.”
- The true believers lifestyle brand is “VT and Chill”
- VT is the Vanguard Total World Stock ETF:
 - A single fund designed to approximate the global investable stock market
 - Covers 10,000+ US and International stocks
 - The basic idea: stop trying to pick winners; buy the haystack

1.2 VT Market Returns

- VT returns come from the stocks it holds:
 - Price changes, weighted by holdings
 - Dividends, distributed quarterly
- Total return combines price growth and dividends.
- Raw prices on the exchange (open, close, high, low)
- Adjusted prices account for dividends.
 - Last year: \$75 price, \$2 dividend
 - Today: \$80 price
 - The dividend makes it as if you bought at \$73
 - Total Return: $100 \times (80 - 73)/73 \approx 9.6\%$

1.3 Quantmod Yahoo Data

```
suppressMessages(library(dplyr))
suppressMessages(library(quantmod))
suppressMessages(library(lubridate))

yahoo_VT = getSymbols('VT', auto.assign = FALSE)
nrow(yahoo_VT)

## [1] 4552

tail(yahoo_VT,5)

##           VT.Open VT.High VT.Low VT.Close VT.Volume
## 2026-07-27  155.48  155.74 153.68   154.45  4989400
## 2026-07-28  153.88  154.82 153.24   154.35  2579800
## 2026-07-29  153.98  154.98 152.13   152.22  4036100
## 2026-07-30  153.98  155.57 153.77   155.45  2716700
## 2026-07-31  155.98  156.25 154.25   155.86  3216200
##           VT.Adjusted
## 2026-07-27        154.45
## 2026-07-28        154.35
## 2026-07-29        152.22
## 2026-07-30        155.45
## 2026-07-31        155.86
```

1.4 VT Price Data

```
class(yahoo_VT)

## [1] "xts" "zoo"

VT = yahoo_VT %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date)) %>%
  rename(price = 'VT.Adjusted') %>%
  select(date, price) %>%
  arrange(date) %>%
  mutate(lag_price = lag(price),
         ror_d = (price/lag_price) - 1)

class(VT)

## [1] "data.frame"
```

1.5 Worst Daily Return

```
VT %>%  
  arrange(ror_d) %>%  
  slice_head(n = 15)
```

##		date	price	lag_price	ror_d
## 1		2008-10-15	22.79921	25.94887	-0.12137967
## 2		2020-03-16	50.69162	57.37014	-0.11641109
## 3		2020-03-12	53.13776	59.19154	-0.10227435
## 4		2008-12-01	19.74520	21.51475	-0.08224834
## 5		2020-03-09	59.56990	64.67337	-0.07891146
## 6		2011-08-08	29.70956	32.07265	-0.07367942
## 7		2009-01-20	19.57765	21.13176	-0.07354386
## 8		2008-10-22	21.50108	23.20231	-0.07332163
## 9		2008-11-20	18.18744	19.47874	-0.06629278
## 10		2008-11-19	19.47874	20.85884	-0.06616407
## 11		2020-03-18	50.19006	53.60412	-0.06369016
## 12		2008-09-29	28.96873	30.92957	-0.06339707
## 13		2008-10-21	23.20231	24.73273	-0.06187835
## 14		2008-11-14	20.88618	22.21163	-0.05967374
## 15		2025-04-04	103.38191	109.93499	-0.05960866

1.6 Random Variables

- View the daily return of VT as a random variable, x .
- A random variable is characterized by its cumulative distribution function (CDF):

$$F(a) = \Pr(x \leq a).$$

- The CDF answers: what fraction of the distribution lies below a ?
- Example:

$$F(0) = \Pr(x \leq 0)$$

is the probability of a non-positive return.

- In the data, estimate this probability with

```
mean(VT$ror_d <= 0).
```

1.7 Empirical CDF

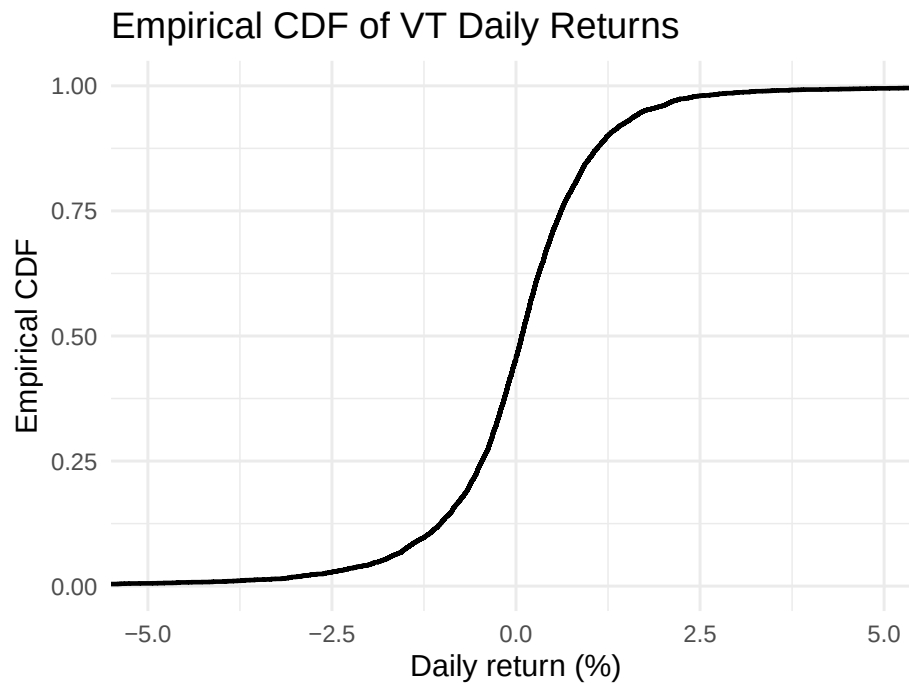
```
mean(VT$ror_d<=0, na.rm = TRUE)
## [1] 0.4594595

ecdf_df = data.frame(x = VT$ror_d) %>%
  filter(!is.na(x)) %>%
  arrange(x) %>%
  mutate(p = row_number()/n())
head(ecdf_df, 5)

##           x           p
## 1 -0.12137967 0.0002197319
## 2 -0.11641109 0.0004394639
## 3 -0.10227435 0.0006591958
## 4 -0.08224834 0.0008789277
## 5 -0.07891146 0.0010986596

library(ggplot2)
ecdf_plot = ggplot(ecdf_df, aes(x = 100*x, y = p)) +
  geom_step(linewidth = 1) +
  coord_cartesian(xlim = c(-5, 5)) +
  labs(x = "Daily return (%)",
       y = "Empirical CDF",
       title = "Empirical CDF of VT Daily Returns") +
  theme_minimal(base_size = 14)
```

1.8



1.9 Quantile Function

- Reversing the CDF gives the quantile function.
- The 0.1-quantile, $q_{0.1}$, satisfies

$$\Pr(x \leq q_{0.1}) = 0.1.$$

That is, 10% of the distribution lies at or below $q_{0.1}$.

- In R:

```
quantile(VT$ror_d, probs = 0.1).
```

1.10 Summarizing a Distribution with Quantiles

- Using quantiles to summarize location, spread, tails
 - In data, use sample quantiles.
- Median, $q_{0.5}$: half the values are below it.

```
quantile(VT$ror_d, probs = 0.5)
median(VT$ror_d)
```

- Minimum and maximum summarize the endpoints.

```
min(VT$ror_d), max(VT$ror_d)
range(VT$ror_d)
```

- The interquartile range summarizes spread:

$$\text{IQR} = q_{0.75} - q_{0.25}.$$

- It measures the width of the middle 50%.

1.11 Quantiles

```
quantile(VT$ror_d, probs = c(0.1, 0.5, 0.9), na.rm = TRUE)
##           10%           50%           90%
## -0.0122397543  0.0007971661  0.0124800276

median(VT$ror_d, na.rm = TRUE)
## [1] 0.0007971661

c(min(VT$ror_d, na.rm = TRUE), max(VT$ror_d, na.rm = TRUE))
## [1] -0.1213797  0.1345332

range(VT$ror_d, na.rm = TRUE)
## [1] -0.1213797  0.1345332

quantile(VT$ror_d,
         probs = c(0.25, 0.75),
         na.rm = TRUE,
         names = FALSE) %>%
  diff()
## [1] 0.01074818

IQR(VT$ror_d, na.rm = TRUE)
## [1] 0.01074818
```

1.12 Parametric Distributions

- So far, we have described returns empirically:
 - empirical CDF, quantiles, median, range, IQR
- Another approach is to assume x follows a parametric distribution:

$$x \sim F_{\theta}.$$

- The parameter vector θ controls the shape of the distribution.
- Example: normal distribution, $\theta = \{\mu, \sigma^2\}$
 - Defined by mean μ and variance σ^2
 - μ controls location; σ controls scale

1.13 Assuming VT is Normally Distributed

- Assume VT daily returns are normally distributed:

$$x \sim N(\mu, \sigma^2), \quad \mu = 0.00042, \quad \sigma = 0.012.$$

- The normal CDF is

$$F(a) = \Pr(x \leq a \mid \mu = 0.00042, \sigma = 0.012).$$

```
pnorm(a, mean = 0.00042, sd = 0.012)
```

- The quantile function satisfies

$$\Pr(x \leq q_\alpha \mid \mu = 0.00042, \sigma = 0.012) = \alpha.$$

`qnorm( $\alpha$ , mean = 0.00042, sd = 0.012)`

1.14 Comparing Returns to a Normal Distribution

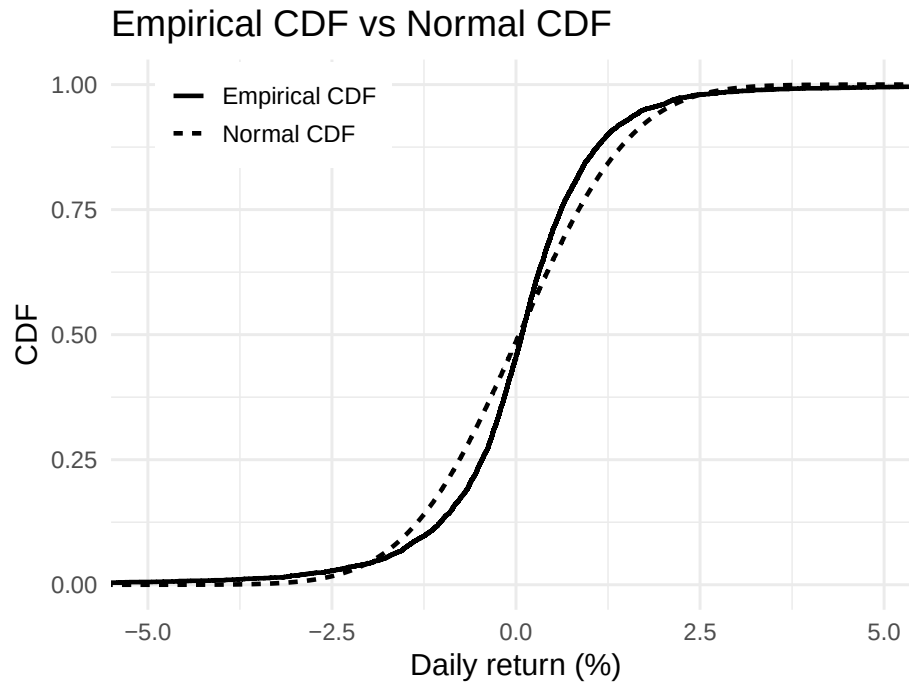
```
ecdf_df2 = ecdf_df %>%
  mutate(normal_cdf = pnorm(x, mean = 0.00042, sd = 0.012))

head(ecdf_df2, 5)

##           x           p    normal_cdf
## 1 -0.12137967 0.0002197319 1.657277e-24
## 2 -0.11641109 0.0004394639 1.059413e-22
## 3 -0.10227435 0.0006591958 5.748974e-18
## 4 -0.08224834 0.0008789277 2.808741e-12
## 5 -0.07891146 0.0010986596 1.909247e-11

ecdf_plot2 = ggplot(ecdf_df2, aes(x = 100*x)) +
  geom_step(aes(y = p, linetype = "Empirical CDF"), linewidth = 1) +
  geom_line(aes(y = normal_cdf, linetype = "Normal CDF"), linewidth = 1) +
  coord_cartesian(xlim = c(-5, 5)) +
  labs(x = "Daily return (%)",
       y = "CDF",
       title = "Empirical CDF vs Normal CDF",
       linetype = NULL) +
  theme_minimal(base_size = 14) +
  theme(legend.position = c(0.2, 0.9),
        legend.background = element_rect(fill = "white", color = NA))
```

1.15



1.16 Common Examples of Parametric Distributions

- Normal: useful for modeling many variables and sampling distributions

`pnorm, qnorm`

- Student- t : useful for small-sample hypothesis tests and fat tails

`pt, qt`

- χ^2 : useful for sampling distributions and hypothesis tests

`pchisq, qchisq`

- F : useful for small-sample tests of multiple restrictions

`pf, qf`

LECTURE 2

Expected Value

2.1 Review

- We looked at daily return of VT as a random variable, x .
- Introduced a random variables cumulative distribution function (CDF):

$$F(a) = \Pr(x \leq a).$$

The CDF answers: what fraction of the distribution lies below a ?

- Reversing the CDF gives the quantile function.
- The 0.1-quantile, $q_{0.1}$, satisfies

$$\Pr(x \leq q_{0.1}) = 0.1.$$

That is, 10% of the distribution lies at or below $q_{0.1}$.

2.2 VT Return Data

```
VT %>%  
  slice_tail(n = 15)  
  
##      date  price lag_price      ror_d  
## 1 2026-08-10 160.97    161.30 -0.0020458886  
## 2 2026-08-11 160.82    160.97 -0.0009318127  
## 3 2026-08-12 161.53    160.82  0.0044148204  
## 4 2026-08-13 162.39    161.53  0.0053240922  
## 5 2026-08-14 162.25    162.39 -0.0008621183  
## 6 2026-08-17 161.83    162.25 -0.0025885866  
## 7 2026-08-18 160.06    161.83 -0.0109374297  
## 8 2026-08-19 160.70    160.06  0.0039984968  
## 9 2026-08-20 159.93    160.70 -0.0047915637  
## 10 2026-08-21 160.77    159.93  0.0052523706  
## 11 2026-08-24 160.10    160.77 -0.0041674327  
## 12 2026-08-25 160.99    160.10  0.0055590216  
## 13 2026-08-26 160.87    160.99 -0.0007454523  
## 14 2026-08-27 161.57    160.87  0.0043514156  
## 15 2026-08-28 161.01    161.57 -0.0034660691
```

2.3 Probability Density Function

- The CDF gives cumulative probability:

$$F(a) = \Pr(x \leq a).$$

- The PDF is the slope of the CDF:

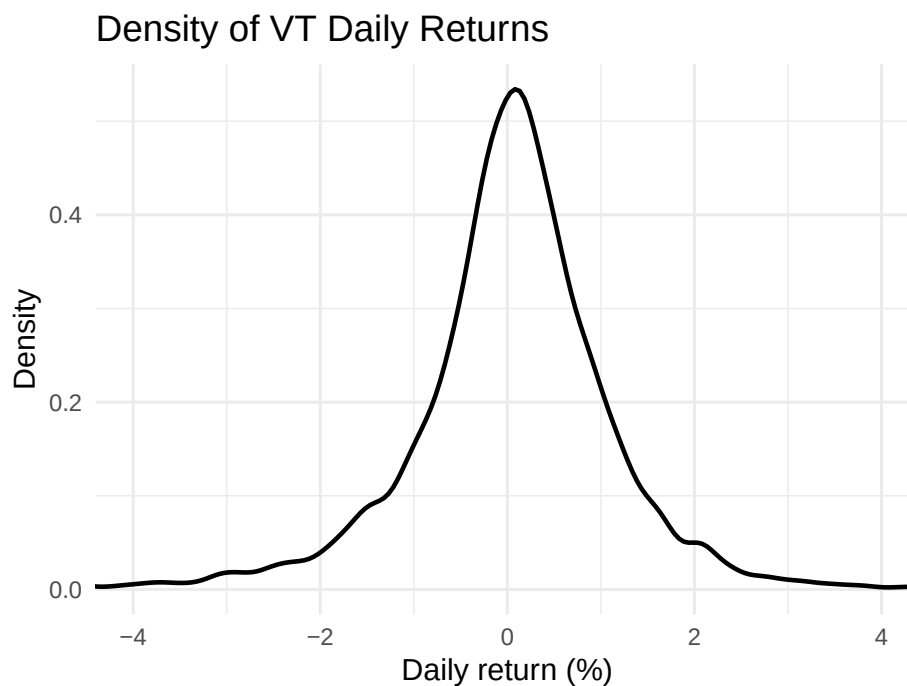
$$f(a) = \frac{dF(a)}{da}.$$

- High density means probability is concentrated close to a .

- Probabilities are areas under the PDF:

$$\Pr(a \leq x \leq b) = \int_a^b f(t) dt = F(b) - F(a).$$

2.4 Density Plot



2.5 Expected Value

- The expected value is the probability-weighted average of a random variable:

$$E(x) = \int x f(x) dx.$$

- It is sometimes called the “typical value,” but this can be misleading.
- With n observations, we estimate it using the sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- In R: `mean(VT$ror_d, na.rm = TRUE)`.

2.6 Properties of Expected Value

- Expected value is linear.

- Adding a constant shifts the expected value:

$$E(x + a) = E(x) + a.$$

- Multiplying by a constant scales the expected value:

$$E(ax) = a E(x).$$

- For two random variables:

$$E(x + y) = E(x) + E(y).$$

- More generally:

$$E(a + bx + cy) = a + b E(x) + c E(y).$$

2.7 Example: Expected Two-Day Return

- Let x_t be today's VT return and x_{t+1} be tomorrow's return.
- The two-day return is

$$R_{2d} = (1 + x_t)(1 + x_{t+1}) - 1.$$

- Expanding,

$$R_{2d} = x_t + x_{t+1} + x_t x_{t+1}.$$

- Therefore, the expected two-day return is

$$E(R_{2d}) = E(x_t) + E(x_{t+1}) + E(x_t x_{t+1}).$$

- The term $E(x_t x_{t+1})$ captures the *dependence* of returns across days.

2.8 Dependence

- There are many ways to describe how one random variable relates to another.
 - These are different notions of *dependence*.
- There are also different ways to describe how one random variable does *not* relate to another.
 - These are different notions of *independence*.
- There are weak forms of independence and stronger forms.
- These distinctions matter because they shape how we analyze data and interpret relationships between variables.

2.9 Covariance

- Covariance measures linear co-movement between two random variables.
- Definition:

$$\text{Cov}(x, y) = E[(x - E(x))(y - E(y))].$$

- Positive: deviations tend to have the same sign.
- Negative: deviations tend to have opposite signs.
- Zero: no systematic linear co-movement.

- Equivalent formula:

$$\text{Cov}(x, y) = E(xy) - E(x)E(y).$$

2.10 A Weak Form of No Dependence

- A weak form of no dependence is zero covariance:

$$\text{Cov}(x, y) = 0.$$

- Since

$$\text{Cov}(x, y) = E(xy) - E(x)E(y),$$

zero covariance implies

$$E(xy) = E(x)E(y).$$

- So

$$E(xy) = E(x)E(y) \quad \Leftrightarrow \quad \text{Cov}(x, y) = 0.$$

- Zero covariance rules out linear co-movement, but it does not rule out all forms of dependence.

2.11 Expected Two-Day Return

- The expected two-day return is

$$E(R_{2d}) = E(x_t) + E(x_{t+1}) + E(x_t x_{t+1}).$$

- Under zero covariance,

$$\text{Cov}(x_t, x_{t+1}) = 0 \quad \Rightarrow \quad E(x_t x_{t+1}) = E(x_t)E(x_{t+1}).$$

- Then the expected two-day return simplifies to

$$E(R_{2d}) = E(x_t) + E(x_{t+1}) + E(x_t)E(x_{t+1}).$$

- It only depends on the expected daily returns

2.12 Constructing $E(R_{2d})$ from One-Day Returns

- If both daily returns have the same distribution,

$$x_t \sim F, \quad x_{t+1} \sim F,$$

then

$$E(x_t) = E(x_{t+1}) = E(x).$$

- Under the assumption $\text{Cov}(x_t, x_{t+1}) = 0$:

$$\begin{aligned} E(R_{2d}) &= E(x_t) + E(x_{t+1}) + E(x_t) E(x_{t+1}) \\ &= E(x) + E(x) + E(x) E(x) \\ &= 2 E(x) + E(x)^2. \end{aligned}$$

- Therefore, we can construct the expected two-day return from the distribution of one-day returns:

$$E(R_{2d}) = (1 + E(x))^2 - 1.$$

2.13 Expected Two-day Return

```
VT = VT %>%
  arrange(date) %>%
  mutate(return_one_day_ago = lag(ror_d),
         price_two_days_ago = lag(price, n = 2),
         return_2d = price / price_two_days_ago - 1)

cov(VT$ror_d, VT$return_one_day_ago, use = "complete.obs")
## [1] -1.779857e-05

mean(VT$return_2d, na.rm = TRUE)
## [1] 0.0008328586

ave_return = mean(VT$ror_d, na.rm = TRUE)
ave_return
## [1] 0.0004245876

(1 + ave_return)^2 - 1
## [1] 0.0008493554
```

2.14 Annual Returns from One-Day Returns

- There are 252 trading days in a year.
- An annual return constructed from one-day returns is

$$R_{252d} = \prod_{s=0}^{251} (1 + x_{t+s}) - 1.$$

- Therefore, the expected annual return is

$$E(R_{252d}) = E \left[\prod_{s=0}^{251} (1 + x_{t+s}) \right] - 1.$$

- Expanding the product creates many expected product terms:

$$E(x_t x_{t+1}), \quad E(x_t x_{t+1} x_{t+2}), \quad E(x_t x_{t+1} x_{t+2} x_{t+3}), \quad \dots$$

2.15 Annual Returns from One-Day Returns

- The expected annual return is

$$E(R_{252d}) = E \left[\prod_{s=0}^{251} (1 + x_{t+s}) \right] - 1.$$

- The many expected product terms in this expression capture dependence across multiple days.
- Suppose all daily returns come from the same distribution:

$$x_{t+s} \sim F, \quad s = 0, 1, 2, \dots, 251.$$

- What assumption about no dependence would simplify the expected annual return to

$$E(R_{252d}) = (1 + E(x))^{252} - 1.$$

2.16 Conditional Distributions

- Assume x_t and x_{t+1} have the same marginal density:

$$x_t \sim f, \quad x_{t+1} \sim f.$$

- Question: what is the distribution of tomorrow's return after a bad day?

$$x_t < -0.01.$$

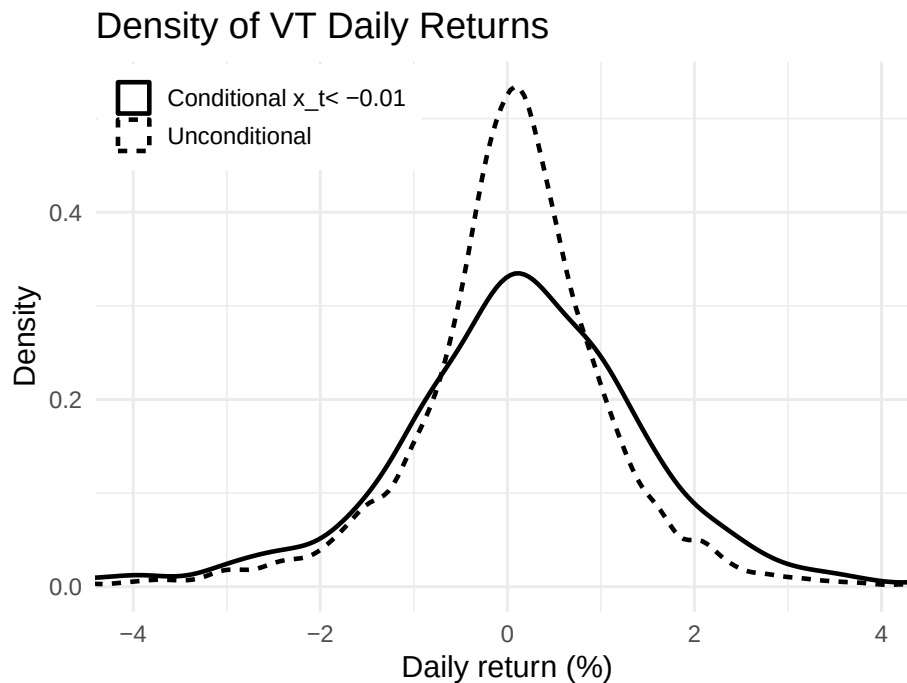
- We represent this with the conditional density:

$$f(x_{t+1} \mid x_t < -0.01).$$

- Compare the unconditional and conditional densities:

$$f(x) \quad \text{vs.} \quad f(x_{t+1} \mid x_t < -0.01).$$

2.17 Comparing the Distributions



2.18 Types of Dependence

- For random variables x and y , if for all values of x ,

$$f(y \mid x) = f(y),$$

then y is *independent* of x .

- Conditioning on x does not change the distribution of y .

- If, for all values of x ,

$$E(y \mid x) = E(y),$$

then y is *mean independent* of x .

- Conditioning on x does not change the mean of y .

- Independence implies mean independence, but mean independence does not imply independence.

2.19 Mean Independence and Covariance

- Mean independence implies

$$E(y \mid x) = E(y).$$

- Use the law of iterated expectations:

$$\begin{aligned} E(xy) &= E_x [E(xy | x)] \\ &= E_x [x E(y | x)] \\ &= E_x [x E(y)] \\ &= E(y) E(x). \end{aligned}$$

- Therefore:

$$\text{Cov}(x, y) = E(xy) - E(x) E(y) = 0.$$

- Mean independence implies zero covariance, but not the other way around.

2.20 Annual Returns from One-Day Returns

- Thus, if daily returns are mean independent of past returns:

$$E(x_{t+s} | x_{t+s-1}, x_{t+s-2}, \dots, x_t) = E(x_{t+s}).$$

- Then expected product terms factor:

$$E(x_t x_{t+1} x_{t+2} \dots) = E(x_t) E(x_{t+1}) E(x_{t+2}) \dots$$

- If daily returns also have the same distribution,

$$E(x_t) = E(x_{t+1}) = \dots = E(x),$$

then the expected annual return simplifies to

$$E(R_{252d}) = (1 + E(x))^{252} - 1.$$

2.21 Expected Annual Return

```
VT = VT %>%
  arrange(date) %>%
  mutate(price_252_days_ago = lag(price, n = 252),
         return_252d = price / price_252_days_ago - 1)

mean(VT$return_252d, na.rm = TRUE)
## [1] 0.1203372

(1 + ave_return)^252 - 1
## [1] 0.1129046
##
## These numbers are close, especially given sampling error.
## Sampling error: variation due to randomness in the observed data.
##
## Appears consistent with mean independence across daily returns.
```

2.22 Summarizing Notions of No Dependence

- From weaker to stronger notions of no dependence.
- *Zero covariance*: no linear co-movement between y and x .

$$\text{Cov}(x, y) = \text{E}[(x - \text{E}(x))(y - \text{E}(y))] = 0.$$

- *Mean independence*: the expected value of y does not change with x .

$$\text{E}(y \mid x) = \text{E}(y).$$

- *Independence*: the distribution of y does not change with x .

$$f(y \mid x) = f(y).$$

LECTURE 3

Variance

3.1 Review

- For a continuous random variable, x , the distribution can be represented by a probability density function:

$$x \sim f(x).$$

- The expected value is

$$E(x) = \int x f(x) dx.$$

- With a sample of n observations,

$$x_1, x_2, \dots, x_n,$$

the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

3.2 Variance

- The mean describes the center of a distribution.
- Variance describes how spread out x is around that center.
- It is the expected squared distance from the mean:

$$\text{Var}(x) = E[(x - E(x))^2].$$

- Define the population mean and variance:

$$\mu = E(x), \quad \sigma^2 = \text{Var}(x).$$

- Then:

$$\sigma^2 = E[(x - \mu)^2].$$

3.3 A Useful Formula for Variance

- Define the population mean: $\mu = E(x)$.
 - μ is a fixed feature of the distribution. It is treated as a constant.
- Using properties of expectation:

$$\begin{aligned} \text{Var}(x) &= E[(x - \mu)^2] \\ &= E[x^2 - 2x\mu + \mu^2] \\ &= E(x^2) - 2\mu E(x) + \mu^2 \\ &= E(x^2) - 2\mu^2 + \mu^2 \\ &= E(x^2) - \mu^2. \end{aligned}$$

- Therefore, variance can also be written as:

$$\text{Var}(x) = E(x^2) - [E(x)]^2.$$

3.4 Estimating Variance

- Estimating $\text{Var}(x) = E[(x - E(x))^2]$ requires an estimate of $E(x)$.
- We estimate $E(x)$ using the sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- The sample variance uses deviations from the estimated center:

$$\widehat{\text{Var}(x)} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- Using $n-1$ instead of n is a correction (discuss later).
- In R: `var(x, na.rm = TRUE)`

3.5 Properties of Variance

- Suppose a is a constant.
- Adding a constant shifts the center but does not change the spread:

$$\text{Var}(x + a) = \text{Var}(x).$$

$$\begin{aligned} \text{Var}(x + a) &= E[(x + a - E(x + a))^2] \\ &= E[(x + a - E(x) - a)^2] \\ &= E[(x - E(x))^2] \\ &= \text{Var}(x). \end{aligned}$$

3.6 Properties of Variance

- Suppose a is a constant.
- Multiplying by a constant rescales the spread:

$$\text{Var}(ax) = a^2 \text{Var}(x).$$

$$\begin{aligned} \text{Var}(ax) &= E[(ax - E(ax))^2] \\ &= E[(ax - aE(x))^2] \\ &= E[a^2(x - E(x))^2] \\ &= a^2 E[(x - E(x))^2] \\ &= a^2 \text{Var}(x). \end{aligned}$$

3.7 Variance of Sum of Two Random Variables

- Expectation is linear:

$$E(x + y) = E(x) + E(y).$$

- Variance is different:

$$\text{Var}(x + y) = \text{Var}(x) + \text{Var}(y) + 2 \text{Cov}(x, y).$$

- Let

$$\tilde{x} = x - E(x), \quad \tilde{y} = y - E(y).$$

$$\begin{aligned} \text{Var}(x + y) &= E[(x + y - E(x + y))^2] \\ &= E[(\tilde{x} + \tilde{y})^2] \\ &= E[\tilde{x}^2 + 2\tilde{x}\tilde{y} + \tilde{y}^2] \\ &= E(\tilde{x}^2) + 2E(\tilde{x}\tilde{y}) + E(\tilde{y}^2) \\ &= \text{Var}(x) + 2 \text{Cov}(x, y) + \text{Var}(y). \end{aligned}$$

3.8 Linear Function to Two Random Variables

- If x and y are random variables and a and b are constants, then

$$E(ax + by) = a E(x) + b E(y).$$

$$\text{Var}(ax + by) = a^2 \text{Var}(x) + b^2 \text{Var}(y) + 2ab \text{Cov}(x, y).$$

3.9 Standard Deviation

- Variance is measured in squared units.
 - If x is measured in dollars, $\text{Var}(x)$ is measured in dollars squared.
- The standard deviation puts spread back in the original units:

$$\text{SD}(x) = \sqrt{\text{Var}(x)}, \quad \sigma = \sqrt{\sigma^2}.$$

- The sample standard deviation is

$$\widehat{\text{SD}}(x) = \sqrt{\widehat{\text{Var}}(x)}.$$

- In R:

$$\text{sd}(x, \text{na.rm} = \text{TRUE}), \text{ or } \text{sqrt}(\text{var}(x, \text{na.rm} = \text{TRUE}))$$

- For roughly normal shaped distributions, most values lie within about $\pm 2\sigma$ of the mean.

3.10 Variance of Returns

- VTI is a total U.S. stock market index fund.
 - Broad exposure to U.S. stocks.
- AGG is a total U.S. bond market index fund.
 - Broad exposure to investment-grade U.S. bonds.
- Variance of returns measures how unpredictable returns are.
 - In finance, this is one way to measure *risk*.
- How do stocks and bonds compare in average return and risk?

3.11 Yearly Returns

```
getSymbols(c('VTI', 'AGG'),
           from = "2004-01-01",
           to = floor_date(Sys.Date(), 'year') - 1)

## [1] "VTI" "AGG"

return_data = merge(VTI %>% Ad() %>% yearlyReturn(),
                    AGG %>% Ad() %>% yearlyReturn()) %>%
  as.data.frame()
names(return_data) = c('VTI', 'AGG')

tail(return_data, 4)

##           VTI           AGG
## 2022-12-30 -0.1952344 -0.13021616
## 2023-12-29  0.2604697  0.05655262
## 2024-12-31  0.2381024  0.01309262
## 2025-12-30  0.1799935  0.07449358

mean(return_data$VTI)
var(return_data$VTI)
sd(return_data$VTI)
```

3.12 Yearly Returns

```
## There are 22 years of annual returns
##
## ----- VTI -----
##
## For VTI, the average annual return is 0.12171
##
## The variance of annual returns is 0.02814
## The standard deviation of annual returns is 0.16775
##
## ----- AGG -----
##
## For AGG, the average annual return is 0.03281
##
## The variance of annual returns is 0.00228
## The standard deviation of annual returns is 0.04777
```

3.13 Risk-Adjusted Return

- VTI has a higher average monthly return.
- AGG has a lower average monthly return, but much lower risk.
- A simple return-to-risk ratio is:

$$\frac{E(r)}{SD(r)}.$$

- This measures average return per unit of volatility.
- A higher value means more average return for each unit of risk.
- What is the ratio for VTI and AGG?

3.14 Portfolio Returns

- Suppose a portfolio invests in VTI and AGG.
- Let VTI_t and AGG_t be their returns in period t .
- An 80/20 portfolio puts 80% in VTI and 20% in AGG:

$$R_t = 0.8VTI_t + 0.2AGG_t.$$

- The expected portfolio return is:

$$E(R_t) = 0.8 E(VTI_t) + 0.2 E(AGG_t).$$

- The variance of the portfolio return is:

$$\begin{aligned} \text{Var}(R_t) &= 0.8^2 \text{Var}(VTI_t) + 0.2^2 \text{Var}(AGG_t) \\ &\quad + 2(0.8)(0.2) \text{Cov}(VTI_t, AGG_t). \end{aligned}$$

3.15 Portfolio Returns

```
return_data = return_data %>%
  mutate(portfolio = 0.8*VTI + 0.2*AGG)
tail(return_data, 3)

##           VTI           AGG portfolio
## 2023-12-29 0.2604697 0.05655262 0.2196863
## 2024-12-31 0.2381024 0.01309262 0.1931004
## 2025-12-30 0.1799935 0.07449358 0.1588935

cov(return_data$VTI, return_data$AGG)
## [1] 0.001494462

(0.8^2)*var(return_data$VTI) + (0.2^2)*var(return_data$AGG) +
  2*0.8*0.2*cov(return_data$VTI, return_data$AGG)
## [1] 0.01857938
##
## The average annual return of portfolio is 0.10393
##
## The variance of annual returns of portfolio is 0.01858
## The standard deviation of annual returns of portfolio is 0.13631
```

3.16 Linear Functions of Random Variables

- A linear function of many random variables can produce many covariance terms.
- If $x_1, \dots, x_n$ are random variables and $a_1, \dots, a_n$ are constants, then

$$\text{Var} \left(\sum_{i=1}^n a_i x_i \right) = \sum_{i=1}^n a_i^2 \text{Var}(x_i) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n a_i a_j \text{Cov}(x_i, x_j).$$

- This simplifies only if the variables are uncorrelated:

$$\text{Var} \left(\sum_{i=1}^n a_i x_i \right) = \sum_{i=1}^n a_i^2 \text{Var}(x_i).$$

- Matrix notation gives a compact way to keep track of all the covariance terms.

3.17 Variance of a Vector Product

- Let x be a column vector of random variables and a be a column vector of constants:

$$x = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}', \quad a = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}'.$$

- The scalar $a'x$ is a weighted sum:

$$a'x = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n.$$

- This gives the sandwich form for the variance

$$\text{Var}\left(\sum_{i=1}^n a_i x_i\right) = \text{Var}(a'x) = a' \text{Var}(x) a.$$

- $\text{Var}(x)$ is an $n \times n$ variance-covariance matrix.

3.18 Variance-Covariance Matrix

- For a vector of random variables,

$$x = [x_1 \quad x_2 \quad \cdots \quad x_n]',$$

the variance of x is

$$\text{Var}(x) = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) & \cdots & \text{Cov}(x_1, x_n) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) & \cdots & \text{Cov}(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(x_n, x_1) & \text{Cov}(x_n, x_2) & \cdots & \text{Var}(x_n) \end{bmatrix}.$$

- The diagonal entries are variances.
- The off-diagonal entries are covariances.

3.19 Portfolio Returns in Vector Form

- Stack the asset returns in a vector:

$$x_t = \begin{bmatrix} \text{VTI}_t \\ \text{AGG}_t \end{bmatrix}.$$

- Stack the portfolio weights in a vector:

$$a = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}.$$

- Portfolio return is a weighted sum:

$$R_t = a'x_t.$$

- Written out:

$$a'x_t = [0.8 \quad 0.2] \begin{bmatrix} \text{VTI}_t \\ \text{AGG}_t \end{bmatrix} = 0.8\text{VTI}_t + 0.2\text{AGG}_t.$$

- The vector product produces one number: the portfolio return.

3.20 Variance of Portfolio Return

- The return vector contains two random variables:

$$x_t = \begin{bmatrix} \text{VTI}_t \\ \text{AGG}_t \end{bmatrix}.$$

- Its variance is a variance-covariance matrix:

$$\begin{aligned} \text{Var}(x_t) &= \begin{bmatrix} \text{Var}(\text{VTI}_t) & \text{Cov}(\text{VTI}_t, \text{AGG}_t) \\ \text{Cov}(\text{AGG}_t, \text{VTI}_t) & \text{Var}(\text{AGG}_t) \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{\text{VTI}}^2 & \sigma_{\text{VTI,AGG}} \\ \sigma_{\text{AGG,VTI}} & \sigma_{\text{AGG}}^2 \end{bmatrix}. \end{aligned}$$

- Since $R_t = a'x_t$, portfolio variance is

$$\text{Var}(R_t) = a' \text{Var}(x_t) a.$$

3.21 Expanding Portfolio Variance

- For the 80/20 portfolio:

$$\begin{aligned} \text{Var}(R_t) &= \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \text{Var}(x_t) \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \begin{bmatrix} \sigma_{\text{VTI}}^2 & \sigma_{\text{VTI,AGG}} \\ \sigma_{\text{VTI,AGG}} & \sigma_{\text{AGG}}^2 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \begin{bmatrix} 0.8\sigma_{\text{VTI}}^2 + 0.2\sigma_{\text{VTI,AGG}} \\ 0.8\sigma_{\text{VTI,AGG}} + 0.2\sigma_{\text{AGG}}^2 \end{bmatrix} \\ &= 0.8(0.8\sigma_{\text{VTI}}^2 + 0.2\sigma_{\text{VTI,AGG}}) \\ &\quad + 0.2(0.8\sigma_{\text{VTI,AGG}} + 0.2\sigma_{\text{AGG}}^2) \\ &= 0.8^2\sigma_{\text{VTI}}^2 + 0.2^2\sigma_{\text{AGG}}^2 + 2(0.8)(0.2)\sigma_{\text{VTI,AGG}}. \end{aligned}$$

- This is the same variance-of-a-sum formula, written in matrix form.

3.22 Variance of Portfolio Returns Using Matrix

```
Vx = cov(return_data[,c('VTI', 'AGG')])
Vx
##           VTI           AGG
## VTI 0.028140453 0.001494462
## AGG 0.001494462 0.002281511

a = rbind(0.8, 0.2)

t(a) %*% Vx %*% a
##           [,1]
## [1,] 0.01857938
```

LECTURE 4

Estimation

4.1 Estimation

- Data come from an underlying process.
- Knowing this process would help guide decisions.
- We typically summarize the process with a finite number of unknown *parameters*.
- Since the process generated the data, we can use the data to learn about the parameters.
 - This is the idea behind *parameter estimation*.
- How much can we learn about the parameter from the data?
- How close is our estimate to the actual parameter?

4.2 Estimation Example

- Suppose we have 36 months of past stock returns.
- We might ask: what will the return be next month?
- The exact return next month is random, so we cannot know it in advance.
- But knowing the distribution of returns would help:

$$x_{t+1} \sim f.$$

- For example, knowing $E(x)$ would tell us the expected return.

4.3 Estimation Example

- Given our sample of 36 returns, a reasonable estimator for $E(x)$ is the sample mean:

$$\bar{x} = \frac{1}{36} \sum_{i=1}^{36} x_i.$$

- $\bar{x}$ will be different from $E(x)$, but hopefully it is close.
- *Key Point:* The data are random, so the estimator is random:

$$x_1, \dots, x_{36} \text{ random} \quad \Rightarrow \quad \bar{x} \text{ random.}$$

- Therefore, $\bar{x}$ has its own distribution.
 - This is the *sampling distribution* of the estimator.
- We evaluate how well $\bar{x}$ recovers $E(x)$ by studying this distribution.

4.4 Sampling Properties of Estimators

- Let $\hat{\mu}$ be an estimator of an unknown parameter μ .
- $\hat{\mu}$ is a random variable because computed from random data.
- We evaluate $\hat{\mu}$ by studying its distribution.
- Desirable properties include:
 - *Unbiasedness*: $E(\hat{\mu}) = \mu$
 - *Consistency*: $\hat{\mu} \rightarrow \mu$ as $n \rightarrow \infty$
 - *Efficiency*: small variance relative to other estimators
 - *Approximate normality*: useful for inference

4.5 Using an Estimator

- Once we have an estimator and know its sampling properties, what can we do?
- Point estimation: $\hat{\mu}$ is a single “best guess” of the unknown parameter.
- Interval estimation: construct a range of plausible values for μ .
 - Constructed from data $\rightarrow$ the interval itself is random.
 - Sampling properties tell us how often this procedure works.
- Hypothesis testing: formally assess claims about the underlying parameter.

4.6 Is the Sample Average a Good Estimator?

- Suppose x_i is a random variable with

$$E(x_i) = \mu, \quad \text{Var}(x_i) = \sigma^2.$$

- We observe a random sample of n observations:

$$x_1, x_2, \dots, x_n.$$

- Each observation is independent and has the same distribution (i.i.d).
- Common estimators for μ and σ^2 are their sample analogs:

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- Based on analogy principle: replace population quantities with sample counterparts.

4.7 Deriving the Sampling Properties

- The estimator is

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- Summation notation is cumbersome. Rewrite as

$$\hat{\mu} = a'x.$$

- Where

$$a = \begin{bmatrix} 1/n \\ 1/n \\ \vdots \\ 1/n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

- a is a vector of constants.
- x is a vector of random variables.

4.8 The Joint Distribution of the Data

- Each x_i is i.i.d.: independent and identically distributed.
- Since each x_i comes from the same distribution, each has the same mean:

$$E(x) = \begin{bmatrix} E(x_1) \\ E(x_2) \\ \vdots \\ E(x_n) \end{bmatrix} = \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix}.$$

4.9 The Joint Distribution of the Data

- The variance-covariance matrix of the data is

$$\text{Var}(x) = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) & \cdots & \text{Cov}(x_1, x_n) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) & \cdots & \text{Cov}(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(x_n, x_1) & \text{Cov}(x_n, x_2) & \cdots & \text{Var}(x_n) \end{bmatrix}.$$

- Each x_i is independent, so covariances are zero.
- Each x_i has the same distribution, so each has the same variance:

$$\text{Var}(x) = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_n.$$

4.10 Expected Value of the Sample Mean

- x is a random vector so $\hat{\mu} = a'x$ is a random variable.

$$\begin{aligned}
 E(\hat{\mu}) &= E(a'x) \\
 &= a'E(x) \\
 &= [1/n \quad 1/n \quad \cdots \quad 1/n] \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix}_{n \times 1} \\
 &= \frac{1}{n}\mu + \frac{1}{n}\mu + \cdots + \frac{1}{n}\mu \\
 &= \mu.
 \end{aligned}$$

- $\hat{\mu}$ is *unbiased*: its expected value equals the parameter we are trying to estimate.

4.11 Variance of the Sample Mean

$$\begin{aligned}
 \text{Var}(\hat{\mu}) &= \text{Var}(a'x) \\
 &= a'\text{Var}(x)a \\
 &= a'(\sigma^2 I_n)a \\
 &= \sigma^2 a'a \\
 &= \sigma^2 [1/n \quad 1/n \quad \cdots \quad 1/n] \begin{bmatrix} 1/n \\ 1/n \\ \vdots \\ 1/n \end{bmatrix} \\
 &= \sigma^2 \left(\frac{1}{n^2} + \frac{1}{n^2} + \cdots + \frac{1}{n^2} \right) \\
 &= \frac{\sigma^2}{n}.
 \end{aligned}$$

- As n gets larger, $\text{Var}(\hat{\mu})$ gets smaller.

4.12 Consistency of the Sample Mean

- The sample mean is unbiased:

$$E(\hat{\mu}) = \mu.$$

- Its variance shrinks with sample size:

$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}.$$

- As $n \rightarrow \infty$,

$$\text{Var}(\hat{\mu}) \rightarrow 0.$$

- Therefore, the sampling distribution of $\hat{\mu}$ collapses around μ .
- This is the intuition for consistency:

$$\hat{\mu} \rightarrow \mu.$$

4.13 The Sample Mean is Normally Distributed

- Two ways to show $\hat{\mu}$ is normally distributed:
- Assume normality of the data:
 - If each $x_i \sim N(\mu, \sigma^2)$, then $\hat{\mu}$ is a sum of normal random variables.
 - Sums of normal random variables are also normal.
- Use the *Central Limit Theorem*:
 - With a large sample, the sample average is approximately normal.
 - This holds even when the underlying data are not normally distributed.

4.14 The Sampling Distribution of $\hat{\mu}$

- Putting it all together:

$$\hat{\mu} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- Standardizing the estimator:

$$\frac{\hat{\mu} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1).$$

- Main challenge: we do not know σ^2 .

- We estimate it using

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- Using $\hat{\sigma}^2$ in place of σ^2 changes the sampling distribution.

4.15 Understanding the Variance of the Estimator

- The sample variance estimates $\text{Var}(x_i) = \sigma^2$:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- Since

$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n},$$

an estimate of the variance of $\hat{\mu}$ is

$$\widehat{\text{Var}(\hat{\mu})} = \frac{\hat{\sigma}^2}{n}.$$

- The estimated standard error is

$$\widehat{\text{SE}}(\hat{\mu}) = \sqrt{\widehat{\text{Var}}(\hat{\mu})} = \sqrt{\frac{\hat{\sigma}^2}{n}}.$$

4.16 Working with the Distribution

- If σ^2 were known, then

$$\frac{\hat{\mu} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1).$$

- But σ^2 is unknown, so we use the estimated standard error:

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})}.$$

- The denominator is now also random.
- This changes the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})} \sim t_{n-1}.$$

4.17 Working with the Distribution

- We have $\hat{\mu}$, consistent and unbiased point estimate.
- We also know the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})} \sim t_{n-1}.$$

- This expression has four components:
 - the estimate, $\hat{\mu}$
 - the unknown parameter, μ
 - the estimated standard error, $\widehat{\text{SE}}(\hat{\mu})$
 - the t_{n-1} distribution
- We know three of them. We invert the expression to learn about the unknown component: μ .

4.18 Estimating the Mean Monthly Returns

```
VTI = getSymbols('VTI',
                  from = floor_date(Sys.Date(), 'month') - years(3),
                  to = floor_date(Sys.Date(), 'month') - 1,
                  auto.assign = FALSE) %>%
  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date))

tail(VTI, 5)

##           date monthly.returns
## 32 2026-03-31   -0.050048376
## 33 2026-04-30    0.104017878
## 34 2026-05-29    0.051838149
## 35 2026-06-30   -0.003853479
## 36 2026-07-30   -0.010188140

nrow(VTI)

## [1] 36
```

4.19 Estimating the Mean Monthly Returns

```
n = nrow(VTI)

ave_return = mean(VTI$monthly.returns)
var_return = var(VTI$monthly.returns)
sd_return = sd(VTI$monthly.returns)

mu_hat = ave_return
mu_hat_var = var_return/n
mu_hat_se = sqrt(mu_hat_var)

##
## The average monthly return is 0.01512
## The standard deviation of monthly returns is 0.03891
## The standard error of the estimate is 0.00649
```

LECTURE 5

Inference

5.1 Reviewing Estimation

- From our sample, we calculate an estimate of an unknown parameter μ .
- Our point estimate $\hat{\mu}$ is a random variable.
- If $\hat{\mu}$ is unbiased and normally distributed, then

$$\hat{\mu} \sim N(\mu, \text{Var}(\hat{\mu})).$$

- We estimate the variance of the estimator using assumptions and data:

$$\widehat{\text{Var}}(\hat{\mu}), \quad \widehat{\text{SE}}(\hat{\mu}) = \sqrt{\widehat{\text{Var}}(\hat{\mu})}.$$

- The finite-sample distribution we use is

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})} \sim t_{n-1}.$$

5.2 Inference

We can use the sampling distribution to:

- 1) Create interval estimates, or confidence intervals.
 - A range of plausible values for μ .
- 2) Conduct hypothesis tests.
 - A formal way to assess claims about a particular value of μ .

5.3 Interval Estimation

- Our estimate may be high or low simply because of sampling randomness.
 - We do not know whether our estimate landed in the lower tail, center, or upper tail.
- But we do know the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})} \sim t_{n-1}.$$

- Most estimates will be within about two standard errors of the true parameter.
- Equivalently, the true parameter will usually be within about two standard errors of the estimate.
- This idea leads to a *confidence interval*.

5.4 Interval Estimation

- $\hat{\mu}$ is our best guess of μ .
 - But $\hat{\mu}$ will generally not exactly equal μ .
- Interval estimation uses the data to identify a range of plausible values:

$$[\text{CI}_{\text{lower}}, \text{CI}_{\text{upper}}].$$

- The confidence level c means:

$$\Pr(\text{CI}_{\text{lower}} \leq \mu \leq \text{CI}_{\text{upper}}) = c.$$

- Interpretation: $(1 - c) \times 100\%$ of the time, the interval fails to cover μ .
 - It is not wide enough to reach μ on one side or the other.

5.5 Interval Estimation

- Interval estimation begins with the sampling distribution.
 - In this case, the t_{n-1} distribution.
- For a given confidence level c , we can find values $\underline{t}$ and $\bar{t}$ such that

$$\Pr(\underline{t} \leq t \leq \bar{t}) = c.$$

- Since the t distribution is symmetric around zero, define one critical value:

$$\kappa = \bar{t} = -\underline{t}.$$

- The expression we work with is

$$\Pr(-\kappa \leq t \leq \kappa) = c.$$

5.6 Finding the Critical Value

- Given the t_{n-1} distribution, for any confidence level c we can find κ such that

$$\Pr(-\kappa \leq t \leq \kappa) = c.$$

- The uncovered probability is split equally across the two tails:

$$\Pr(t > \kappa) = \frac{1 - c}{2}, \quad \Pr(t < -\kappa) = \frac{1 - c}{2}.$$

- We calculate κ using the inverse CDF:

$$\text{crit_value} = -\text{qt}((1 - \text{conf_lvl})/2, \text{df}=\text{n}-1).$$

5.7 Interval Estimation Math

- We start with the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})} \sim t_{n-1}.$$

- Given confidence level c , choose κ such that

$$\Pr\left(-\kappa \leq \frac{\hat{\mu} - \mu}{\widehat{\text{SE}}(\hat{\mu})} \leq \kappa\right) = c.$$

- Rearranging gives:

$$\Pr\left(\hat{\mu} - \kappa \widehat{\text{SE}}(\hat{\mu}) \leq \mu \leq \hat{\mu} + \kappa \widehat{\text{SE}}(\hat{\mu})\right) = c.$$

- Therefore, the confidence interval is

$$\left[\hat{\mu} - \kappa \widehat{\text{SE}}(\hat{\mu}), \quad \hat{\mu} + \kappa \widehat{\text{SE}}(\hat{\mu})\right].$$

5.8 Steps to Interval Estimation

- (1) Choose confidence level c .

– Example: $c = 0.95$, a 95% confidence interval.

- (2) Choose critical value κ from the t_{n-1} distribution:

$$\Pr(t > \kappa) = \frac{1 - c}{2}.$$

- (3) Construct the interval:

$$\left[\hat{\mu} - \kappa \widehat{\text{SE}}(\hat{\mu}), \quad \hat{\mu} + \kappa \widehat{\text{SE}}(\hat{\mu})\right].$$

- Tradeoff:

- Higher confidence $\Rightarrow$ larger $\kappa \Rightarrow$ wider interval.
- Wider intervals are more likely to cover μ , but less precise.

5.9 Confidence Interval for Mean Monthly Return

```
VTI = getSymbols('VTI',
                 from = floor_date(Sys.Date(), 'month') - years(3),
                 to = floor_date(Sys.Date(), 'month') - 1,
                 auto.assign = FALSE) %>%

  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date))
n = nrow(VTI)
mu_hat = mean(VTI$monthly.returns)
mu_hat_se = sqrt(var(VTI$monthly.returns)/n)

##
## The estimated expected monthly return is 0.01512
##
## The standard error of the estimate is 0.00649
```

5.10 Confidence Interval for Mean Monthly Return

```
crit_val_95 = -qt((1 - 0.95)/2, df = n - 1)
mu_hat_lb_95 = mu_hat - crit_val_95*mu_hat_se
mu_hat_ub_95 = mu_hat + crit_val_95*mu_hat_se

crit_val_99 = -qt((1 - 0.99)/2, df = n - 1)
mu_hat_lb_99 = mu_hat - crit_val_99*mu_hat_se
mu_hat_ub_99 = mu_hat + crit_val_99*mu_hat_se

##
## The critical value for 95% CI is 2.03011
##
## The lower bound 95% CI is 0.00196
##
## The upper bound 95% CI is 0.02829
##
## The critical value for 99% CI is 2.72381
##
## The lower bound 99% CI is -0.00254
##
## The upper bound 99% CI is 0.03279
```

5.11 Hypothesis Testing

- Consider the *null hypothesis*: $\mu = \mu_0$.
- If the null is true, the sampling distribution of $\hat{\mu}$ is

$$\hat{\mu} \sim N(\mu_0, \text{Var}(\hat{\mu})).$$

- We expect most values of $\hat{\mu}$ to be close to μ_0 .

- If $\hat{\mu}$ is far from μ_0 , this could happen because:
 - 1) an unusual sample produced an unusual estimate.
 - 2) the null hypothesis is wrong.
- Hypothesis testing formalizes when (1) becomes unlikely enough to reject the null.

5.12 Hypothesis Testing of a Single Parameter

- In a two-sided test, we state the null hypothesis:

$$H_0 : \mu = \mu_0.$$

- The alternative hypothesis is:

$$H_1 : \mu \neq \mu_0.$$

- Hypothesis testing asks:
 - If the null is true, how likely are we to observe our estimate, or one more extreme?
 - This probability is called the *p-value*.
- If the p-value is below a chosen threshold, such as 0.05 or 0.01, we reject the null.
 - *Rejecting the null* means the data provide evidence that $\mu \neq \mu_0$.

5.13 T-Test

- From the sample, we estimate $\hat{\mu}$ and $\widehat{\text{SE}}(\hat{\mu})$.
- Consider the hypotheses:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0.$$

- Calculate the t-statistic:

$$t = \frac{\hat{\mu} - \mu_0}{\widehat{\text{SE}}(\hat{\mu})}.$$

- The t-statistic measures how far $\hat{\mu}$ is from μ_0 , adjusted for sampling uncertainty.
- If the null is true, this statistic follows a t_{n-1} distribution, centered at zero.
- Large absolute values of t are evidence against the null.

5.14 P-Value

- For every hypothesis test, we can calculate a *p-value*.
 - If the null is true, this is the probability of observing a test statistic at least as extreme as ours.
- Let $F_{t_{n-1}}(a)$ denote the CDF of a t distribution with $n - 1$ degrees of freedom.

- Since the t-statistic follows this distribution under the null, for a two-sided test:

$$\text{p-value} = 2 \times [1 - F_{t_{n-1}}(|t|)] .$$

- In R:

```
2 * (1 - pt(abs(tstat), df = n - 1))
```

5.15 Significance Level

- To conclude the hypothesis test, we choose a significance level, α .
- α is the probability of rejecting the null when the null is true.
 - $\alpha = 0.05$: 1/20 times reject a true null.
 - $\alpha = 0.01$: 1/100 times reject a true null.
- Decision rule:
 - If $\text{p-value} < \alpha$, *reject the null* at significance level α .
 - * The data provide evidence against the null.
 - If $\text{p-value} \geq \alpha$, *fail to reject the null*.

5.16 Test Mean Monthly Return VTI is 0

```
mu_null = 0
t_stat = (mu_hat - mu_null)/mu_hat_se
p_val = 2*(1 - pt(abs(t_stat), df = n - 1))

##
## The t-statistic is 2.33169
##
## The p-value is 0.02560
##
## With our 36 observations
##
## Reject null that the month return to VTI is zero at the 5% level.
##
## However, fail to reject at the 1% level.
```

5.17 Failing to Reject the Null

- We fail to reject the null when t is close to zero.:

$$t = \frac{\hat{\mu} - \mu_0}{\widehat{SE}(\hat{\mu})}.$$

- This can happen for two reasons:
 - 1) $\hat{\mu}$ is close to μ_0 .
 - 2) $\widehat{SE}(\hat{\mu})$ is large, so the estimate is imprecise.

- Failing to reject should be interpreted cautiously.
 - If $\widehat{SE(\hat{\mu})}$ is small, failing to reject may provide some support for the null.
 - If $\widehat{SE(\hat{\mu})}$ is large, the test may simply be inconclusive.

5.18 Critical Value

- We can conduct a hypothesis test at significance level α using a critical value.
- For a two-sided test with test statistic from a t_{n-1} distribution, choose κ such that

$$\Pr(t > \kappa) = \frac{\alpha}{2}.$$

- The rejection region is

$$t < -\kappa \quad \text{or} \quad t > \kappa$$

- Decision rule:
 - Reject the null if $|t| > \kappa$.
 - Fail to reject the null if $|t| \leq \kappa$.
- In R: `crit_value = -qt(alpha / 2, df = n - 1)`

5.19 Hypothesis Tests and Confidence Intervals

- The critical value for a two-sided test at significance level α is

$$\text{crit_value} = -\text{qt}(\alpha / 2, \text{df} = n - 1).$$

- This is the same critical value used for a $(1 - \alpha)$ confidence interval.
- Therefore, we can use the confidence interval to conduct the hypothesis test.
- Example: given a 95% confidence interval,
 - If μ_0 is inside the confidence interval, fail to reject $H_0 : \mu = \mu_0$ at the 5% level.
 - If μ_0 is outside the confidence interval, reject $H_0 : \mu = \mu_0$ at the 5% level.

LECTURE 6

Simulation Methods

6.1 Simulation Methods

- Many statistical questions are easy to ask but hard to answer analytically. The algebra is difficult.
 - Example: what is the sampling distribution of my estimator?
- The difficulty is that we usually observe only one random sample.
 - If we observed many random samples, the question would be easy to answer.
- Simulation methods repeatedly create artificial datasets to answer questions.
 - These methods play a growing role with advances in modern computing power.

6.2 Correlation

- Correlation is standardized covariance:

$$\rho = \text{Cor}(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x) \text{Var}(y)}}.$$

- Measures linear co-movement in SD units.
 - It does not depend on the units of x or y .
- If $\text{Cor}(x, y) = 1$, one SD higher in x corresponds to one SD higher in y .
- Correlation takes values between -1 and 1 .
 - 1 : perfect positive linear relationship
 - 0 : no linear relationship
 - -1 : perfect negative linear relationship

6.3 Estimating Correlation

- Correlations are straightforward to estimate from the sample:

$$\hat{\rho} = \widehat{\text{Cor}(x, y)} = \frac{\widehat{\text{Cov}(x, y)}}{\sqrt{\widehat{\text{Var}(x)} \widehat{\text{Var}(y)}}.$$

- But deriving the standard error,

$$\text{SE} \left(\widehat{\text{Cor}(x, y)} \right),$$

is less straightforward.

- The statistic depends on a complicated nonlinear function of the sample.
- This type of problem is a natural candidate for bootstrapping.

6.4 Returns to Stocks and Bonds

```
getSymbols(c('VTI', 'AGG'),
           from = "2004-01-01",
           to = floor_date(Sys.Date(), 'month') - 1)

## [1] "VTI" "AGG"

VTI = VTI %>%
  Ad() %>%
  monthlyReturn()
names(VTI) = 'VTI'

AGG = AGG %>%
  Ad() %>%
  monthlyReturn()
names(AGG) = 'AGG'

ETF_data = merge(VTI, AGG) %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date))

##           date          VTI          AGG
## 269 2026-05-29  0.051838149  0.002936027
## 270 2026-06-30 -0.003853479  0.002542450
## 271 2026-07-30 -0.010188140 -0.010430929
##
## Data has 271 observations of monthly returns
```

6.5 Correlation of Returns of Stocks and Bonds

```
cor(ETF_data$VTI, ETF_data$AGG)
## [1] 0.2298297

cov(ETF_data$VTI, ETF_data$AGG)/
  sqrt(var(ETF_data$VTI)*var(ETF_data$AGG))
## [1] 0.2298297

cov(ETF_data$VTI, ETF_data$AGG)/
  (sd(ETF_data$VTI)*sd(ETF_data$AGG))
## [1] 0.2298297
```

6.6 Bootstrapping a Standard Error

- $\widehat{SE}(\hat{\rho})$ measures how much $\hat{\rho}$ varies due to random sampling.
- If we had B samples, we could calculate B estimates:

$$\hat{\rho}_{(1)}^*, \hat{\rho}_{(2)}^*, \dots, \hat{\rho}_{(B)}^*.$$

- Then we could estimate the standard error directly:

$$\widehat{\text{SE}}(\hat{\rho}) = \sqrt{\frac{1}{B-1} \sum_{s=1}^B \left(\hat{\rho}_{(s)}^* - \bar{\hat{\rho}}^* \right)^2}.$$

- Bootstrapping calculates the standard error by creating artificial samples by resampling from the original data

6.7 Comments on Bootstrapping

- Bootstrap logic: our data are a random sample from the population.
 - Resampling from the data approximates sampling from the population.
- Bootstrap standard errors can differ from analytical formulas.
 - They may be more credible when analytical assumptions are questionable.
- There are many bootstrap strategies.
 - Non-parametric bootstrap, parametric bootstrap, wild bootstrap, block bootstrap, etc.

6.8 Non-Parametric Bootstrapping

- We have n pairs of annual returns for VTI and AGG:

$$(\text{VTI}_1, \text{AGG}_1), (\text{VTI}_2, \text{AGG}_2), \dots, (\text{VTI}_n, \text{AGG}_n).$$

- 1) Create B samples by sampling n pairs *with replacement*.
 - Larger B gives a more accurate bootstrap approximation.
- 2) For each sample, calculate $\hat{\rho}_{(s)}^*$.
- 3) Estimate the standard error:

$$\widehat{\text{SE}}(\hat{\rho}) = \text{SD} \left(\hat{\rho}_{(s)}^* \right).$$

6.9 Bootstrapping a Standard Error: Expanding

```
rho_hat = cor(ETF_data$VTI, ETF_data$AGG)

n_boots = 5000
n_obs = nrow(ETF_data)
boot_data = ETF_data %>%
  slice_sample(n = n_boots*n_obs, replace = TRUE) %>%
  mutate(boot_idx = rep(1:n_boots, each = n_obs)) %>%
  group_by(boot_idx) %>%
  summarize(rho_hat = cor(VTI, AGG))

rho_hat_se = sd(boot_data$rho_hat)

##
## The standard error of the estimate is 0.07045
```

6.10 Bootstrapping a Standard Error: Loop

```
n_boots = 5000
n_obs = nrow(ETF_data)
boot_data = vector(mode = 'numeric', length = n_boots)
for (s in 1:n_boots){
  samp_idx = sample(n_obs, n_obs, replace = TRUE)
  boot_data[s] = cor(ETF_data$VTI[samp_idx], ETF_data$AGG[samp_idx])
}
rho_hat_se = sd(boot_data)

##
## The standard error of the estimate is 0.07011
```

6.11 Correlation of Returns of Stocks and Bonds

```
rho_null = 0
tstat = (rho_hat - rho_null)/rho_hat_se
pval = 2*(1 - pt(abs(tstat), df = n_obs - 1))

crit_val = -qt((1-0.95)/2, df = n_obs - 1)

CI95 = rho_hat + c(-1, 1)*crit_val*rho_hat_se

##
## The number of observations is 271.
##
## The estimated correlation is 0.22983.
##
## The standard error of the estimate is 0.07011.
##
## Testing H0: rho = 0.00, the t-statistic is 3.278.
##
## The p-value for the test is 0.0012.
##
## The critical value for a 95% CI is 1.969.
##
## The 95% confidence interval is [0.09180, 0.36786].
```

6.12 Monte Carlo Simulation

- We have about 20 years of return data.
- Suppose we want to study ten-year returns.
 - For example: mean, variance, probability of losing money.
- With only 20 years of data, we have only about two non-overlapping ten-year returns.
 - Not enough to estimate these quantities reliably.
- Overlapping ten-year returns create more observations.
 - But the observations are highly dependent, which creates complications

6.13 Monte Carlo Simulation

- If we had a sample of ten-year returns,

$$R_{10Y,1}, R_{10Y,2}, R_{10Y,3}, \dots, R_{10Y,n},$$

estimating the mean and variance would be straightforward.

- Monte Carlo simulation uses the data to create many artificial ten-year return histories.
- Treat the simulated ten-year returns like a sample and calculate sample moments.

6.14 Review of Estimating Ten-Year Returns

- Let x_t be monthly returns.
- A ten-year return compounds 120 monthly returns:

$$R_{10Y} = \prod_{j=0}^{120-1} (1 + x_{t+j}) - 1.$$

- Assume monthly returns have a common mean $E(x_t)$, estimated by the sample average:

$$\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t.$$

- We showed that if monthly returns are uncorrelated, the expected ten-year return can be estimated as

$$\bar{R}_{10Y} = (1 + \bar{x})^{120} - 1.$$

6.15 Monte Carlo Simulation

- The analytical approach gives a simple estimate:

$$\bar{R}_{10Y} = (1 + \bar{x})^{120} - 1.$$

- Monte Carlo simulation takes a different approach.
- Simulate many artificial ten-year return histories.
 - Each history contains 120 monthly returns.
 - Assume monthly returns are uncorrelated.
- For each simulation, calculate the ten-year return:

$$R_{10Y}^{(s)} = \prod_{j=0}^{12 \times 10 - 1} (1 + x_j^{(s)}) - 1.$$

- Average across simulations to estimate the expected ten-year return.

6.16 Simulated 10-Year Return

```
VTI_mean = mean(ETF_data$VTI)
VTI_10Y_closed = (1 + VTI_mean)^(120) - 1

n_mc = 10000
VTI_MC = ETF_data %>%
  slice_sample(n = n_mc*120, replace = TRUE) %>%
  mutate(mc_idx = rep(1:n_mc, each = 120)) %>%
  group_by(mc_idx) %>%
  summarize(R10Y = prod(1 + VTI) - 1)
VTI_10Y_sim = mean(VTI_MC$R10Y)

##
## The close-form analytical 10-year VTI return is 214.376%.
##
## The simulated 10-year VTI return is 213.400%.
```

6.17 Ten-Year Return Variance

- Expected ten-year returns are manageable analytically.
- Variance is harder because returns are compounded:

$$R_{10Y} = \prod_{j=0}^{119} (1 + x_{t+j}) - 1.$$

- Expanding creates $2^{120} - 1$ product terms.
 - monthly returns: $x_t, x_{t+1}, \dots$
 - two-month products: $x_t x_{t+1}, x_t x_{t+2}, \dots$
 - all the way up to $x_t x_{t+1} \cdots x_{t+119}$
- Monte Carlo avoids managing this formula directly.

6.18 Simulated 10-Year Return VTI

```
VTI_10Y_mean = mean(VTI_MC$R10Y)
VTI_10Y_sd = sd(VTI_MC$R10Y)
VTI_10Y_rar = VTI_10Y_mean / VTI_10Y_sd

##
##
## VTI 10-year return
##
## Average return: 213.40%
##
## Standard deviation: 156.51%
##
## Risk-adjusted return: 1.3635
```

6.19 Simulated 10-Year Return 80/20 Portfolio

```
n_mc = 10000
Portfolio_MC = ETF_data %>%
  slice_sample(n = n_mc*120, replace = TRUE) %>%
  mutate(mc_idx = rep(1:n_mc, each = 120)) %>%
  group_by(mc_idx) %>%
  summarize(R10Y = prod(1 + 0.8 * VTI + 0.2 * AGG) - 1)
Portfolio_10Y_mean = mean(Portfolio_MC$R10Y)
Portfolio_10Y_sd = sd(Portfolio_MC$R10Y)
Portfolio_10Y_rar = Portfolio_10Y_mean / Portfolio_10Y_sd

##
##
## 80/20 Portfolio 10-year return
##
## Average return: 166.02%
##
## Standard deviation: 105.55%
##
## Risk-adjusted return: 1.5729
```

LECTURE 7

Prediction and OLS

7.1 Prediction

- We have an outcome y we want to predict.
- y is a random variable that may depend on observed and unobserved factors.
- For observation i , a prediction function maps observed information into a prediction:

$$\hat{y}_i = g(x_i).$$

- The prediction error is

$$e_i = y_i - g(x_i).$$

- We evaluate predictions using squared error:

$$e_i^2 = (y_i - \hat{y}_i)^2 = (y_i - g(x_i))^2.$$

7.2 Estimating a Prediction Function

- We observe n pairs:

$$(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n).$$

- Choose a prediction function with unknown parameters:

$$g(x_i).$$

- Estimate the parameters by minimizing the sum of squared errors:

$$\text{SSE} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - g(x_i))^2.$$

7.3 Ordinary Least Squares

- A common prediction function is linear:

$$g(x_i) = b_1 + b_2 x_i.$$

- Choose b_1 and b_2 to minimize the sum of squared errors:

$$\text{SSE} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2.$$

- This estimation method is called *ordinary least squares*.
- OLS provides the best linear predictor.

7.4 OLS with Multiple Predictors

- In practice, we use many predictors to predict y .
- Let x_i be a $K \times 1$ vector:

$$x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iK} \end{bmatrix}.$$

- The prediction function becomes

$$g(x_i) = x_i' b,$$

where b is a $K \times 1$ vector of parameters.

7.5 Understanding the Prediction

- The prediction is a vector product:

$$\begin{aligned} \hat{y}_i = g(x_i) &= \begin{bmatrix} x_{i1} & x_{i2} & \cdots & x_{iK} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_K \end{bmatrix} \\ &= x_{i1}b_1 + x_{i2}b_2 + \cdots + x_{iK}b_K. \end{aligned}$$

- Typically, define $x_{i1} = 1$ as the constant term.
- Then b_1 is the intercept and the prediction when all other predictors equal zero:

$$\hat{y}_i = b_1 \quad \text{when } x_{i2} = \cdots = x_{iK} = 0.$$

7.6 The OLS Estimator

- Choose b to minimize the sum of squared errors:

$$b = \underset{b}{\operatorname{argmin}} \sum_{i=1}^n (y_i - x_i' b)^2.$$

- Since

$$\begin{aligned} x_i' b &= x_{i1}b_1 + x_{i2}b_2 + \cdots + x_{iK}b_K, \\ \text{SSE} &= \sum_{i=1}^n (y_i - x_{i1}b_1 - x_{i2}b_2 - \cdots - x_{iK}b_K)^2. \end{aligned}$$

- Take K partial derivatives and set them equal to zero:

$$\frac{\partial \text{SSE}}{\partial b_k} = -2 \sum_{i=1}^n x_{ik} (y_i - x_i' b) = 0, \quad k = 1, \dots, K.$$

7.7 The OLS Solution

- The OLS estimator chooses b to satisfy

$$\frac{\partial \text{SSE}}{\partial b_k} = -2 \sum_{i=1}^n x_{ik} (y_i - x'_i b) = 0, \quad k = 1, \dots, K.$$

- Recall that the error is

$$e_i = y_i - x'_i b.$$

- Therefore, the first-order conditions imply

$$\sum_{i=1}^n x_{ik} e_i = 0, \quad k = 1, \dots, K.$$

7.8 Vectorizing the OLS Solution

- The K first-order conditions are

$$\sum_{i=1}^n x_{i1} e_i = 0, \quad \sum_{i=1}^n x_{i2} e_i = 0, \quad \dots \quad \sum_{i=1}^n x_{iK} e_i = 0.$$

- Define $\mathbf{x}_k$ as the $n \times 1$ vector containing predictor k :

$$\mathbf{x}_k = \begin{bmatrix} x_{1k} \\ x_{2k} \\ \vdots \\ x_{nk} \end{bmatrix}, \quad e = \begin{bmatrix} y_1 - x'_1 b \\ y_2 - x'_2 b \\ \vdots \\ y_n - x'_n b \end{bmatrix} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}.$$

- Each condition can be written as

$$\mathbf{x}'_k e = 0, \quad k = 1, \dots, K.$$

7.9 In Matrix Form

- Combine the K predictor vectors column-wise to create an $n \times K$ matrix:

$$X = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_K].$$

- Y is the $n \times 1$ vector of outcomes:

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \hat{Y} = \begin{bmatrix} x'_1 b \\ x'_2 b \\ \vdots \\ x'_n b \end{bmatrix} = Xb.$$

- The error vector is

$$e = Y - \hat{Y} = Y - Xb.$$

- The OLS first-order conditions are

$$X'e = X'(Y - \hat{Y}) = X'(Y - Xb) = \mathbf{0}_{K \times 1}.$$

7.10 The OLS Normal Equations

- The first-order conditions are

$$X'(Y - Xb) = \mathbf{0}_{K \times 1}.$$

- Solve the system of equations:

$$\begin{aligned} \mathbf{0} &= X'(Y - Xb) \\ &= X'Y - X'Xb \\ X'Xb &= X'Y \\ (X'X)^{-1}(X'X)b &= (X'X)^{-1}X'Y \\ I_{K \times K}b &= (X'X)^{-1}X'Y \\ \boxed{b} &= (X'X)^{-1}X'Y \end{aligned}$$

7.11 OLS Example: MLB Data

- Analyze the 2023 season for 30 MLB teams.
- Let total season wins for team i be

$$\text{Wins}_i.$$

- Estimate the prediction function using OLS:

$$\widehat{\text{Wins}}_i = b_1 + b_2 R_i + b_3 \text{RA}_i.$$

- R_i is runs scored by team i .
- RA_i is runs allowed by team i .

7.12 OLS with MLB Data

- Build the outcome vector:

$$Y = \begin{bmatrix} \text{Wins}_1 \\ \text{Wins}_2 \\ \vdots \\ \text{Wins}_n \end{bmatrix}_{n \times 1}.$$

- Build the design matrix:

$$X = \begin{bmatrix} 1 & R_1 & \text{RA}_1 \\ 1 & R_2 & \text{RA}_2 \\ \vdots & \vdots & \vdots \\ 1 & R_n & \text{RA}_n \end{bmatrix}_{n \times 3}.$$

- Estimate the OLS coefficients:

$$b = (X'X)^{-1}X'Y.$$

7.13 Lahman Library

```
suppressMessages(library(dplyr))
suppressMessages(library(Lahman))

mlb_2023 <- Teams %>%
  filter(yearID == 2023) %>%
  select(teamID, name, W, R, RA)

##
## Data on 30 MLB teams in 2023
##
##      teamID              name  W   R  RA
## 21    PHI Philadelphia Phillies 90 796 715
## 22    PIT   Pittsburgh Pirates 76 692 790
## 23    SDN   San Diego Padres 82 752 648
## 24    SEA   Seattle Mariners 88 758 659
## 25    SFN San Francisco Giants 79 674 719
## 26    SLN   St. Louis Cardinals 71 719 829
## 27    TBA   Tampa Bay Rays 99 860 665
## 28    TEX   Texas Rangers 90 881 716
## 29    TOR   Toronto Blue Jays 89 746 671
## 30    WAS   Washington Nationals 71 700 845
```

7.14 OLS Estimates

```
n = nrow(mlb_2023)
Y = matrix(mlb_2023$W, ncol = 1)
X = as.matrix(cbind(rep(1, n),
                    mlb_2023$R,
                    mlb_2023$RA))

b = solve( t(X) %*% X) %*% (t(X) %*% Y)

##           b1           b2           b3
## 89.55410905  0.08284065 -0.09428070
```

7.15 Decomposing the Total Variation in y

- Given a prediction, decompose the outcome as

$$y_i = \underbrace{\hat{y}_i}_{\text{prediction}} + \underbrace{e_i}_{\text{prediction error}}.$$

- The Sum of Squares Total (SST) describes the total variation in y :

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

- The Sum of Squared Errors (SSE) describes the variation in y unexplained by the prediction:

$$\text{SSE} = \sum_{i=1}^n e_i^2.$$

7.16 Goodness of Fit: R^2

- R^2 characterizes how well we fit or predict the outcome variable.
- R^2 is the fraction of sample variation in y explained by the prediction $\hat{y}$.
- Since SSE/SST is the fraction unexplained,

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} \in [0, 1].$$

- $R^2 = 0 \rightarrow 0\%$ of the variation in y is explained by $\hat{y}$.
- $R^2 = 1 \rightarrow 100\%$ of the variation in y is explained by $\hat{y}$.

7.17 R-Squared

```
err = Y - X*%b
SSE = sum(err^2)
SST = sum((Y - mean(Y))^2)
r_squared = 1 - SSE/SST
```

##	SSE	SST	R-squared
##	538.5235663	4994.0000000	0.8921659

7.18 An Alternative MLB Predictor

- b_3 appears very similar to $-b_2$.
- Estimate the prediction function under the assumption $b_3 = -b_2$:

$$\begin{aligned}\widehat{\text{Wins}}_i &= b_1 + b_2 R_i + b_3 \text{RA}_i \\ &= b_1 + b_2 R_i - b_2 \text{RA}_i \\ &= b_1 + b_2 (R_i - \text{RA}_i) \\ &= b_1 + b_2 \text{RD}_i,\end{aligned}$$

where RD_i is run differential.

- Do run levels matter, or only run differential?
 - Compare the R^2 between the two prediction functions.

7.19 Comparing the R-Squared

```
X_sm = as.matrix(cbind(rep(1, n),
                        mlb_2023$R - mlb_2023$RA))
b_sm = solve( t(X_sm) %*% X_sm) %*% (t(X_sm) %*% Y)
err_sm = Y - X_sm%*%b_sm

r_squared_sm = 1 - sum(err_sm^2)/SST

##          b1          b2
## 81.00000000  0.08852526
##
## Original R-squared 0.89217
##
## R-squared from smaller prediction equation 0.89071
```